# Ganesh Iyer

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Google Scholar Portfolio

## Education

Sep 2025–2027 **ETH Zürich**, *Master of Science*,
Robotics, Systems and Control

GPA **5.72/6**

Nov 2021– May 2025 **Indian Institute of Technology Bombay**, *Bachelor of Technology*,
Mechanical Engineering (with Honors) specializing in Systems and Controls,
Minor in Artificial Intelligence and Data Science

GPA **9.87/10**

## Publications

Conference Paper: Lin Yang, **Ganesh Iyer**, Baichuan Lou, Sri Harsha Turlapati, Chen Lv, Domenico Campolo, **"Path Planning in Complex Environments with Superquadrics and Voronoi-Based Orientation"**, IEEE Conference on Robotics and Biomimetics 2025

## Skills

Languages: Python, MATLAB, C++

Software: ROS2, NVIDIA Isaac Lab/Sim/Gym, Git, GitHub, Docker, LaTeX

Libraries: OpenCV, Pandas, Pytorch, NumPy, Scikit Learn

## Internships

June 2025 – July 2025 **Reinforcement Learning for Quadruped Robot Locomotion**
xTerra Robotics, India

- Researched **quadruped locomotion control**, combining model- and learning-based methods
- Trained RL policy for Svan M2 via **guided policy search** with trajectory tracking rewards, using data from **trajectory planner** generated offline, resulting in accelerated training
- Used PPO training in NVIDIA Isaac Gym with domain randomization for Sim2Real, validated in **MuJoCo**
- Trained NN-based **state estimator** for base linear, angular velocity and base height estimation from sensor data buffer, required by the RL Policy
- Deployed on Svan M2 **hardware**, achieving **trot gait** (0.7 m/s forward, 0.5 m/s lateral, 1 rad/s yaw)
- Video link of training, simulation and hardware results can be found here

May 2024 – Jul 2024 **Voronoi Diagram Based Path Planning for Superquadric Obstacles**
Global Connect Fellowship, Nanyang Technological University | Guide: Prof Domenico Campolo

- Conducted a comprehensive survey on **superquadrics** and their role in obstacle-based **path planning**.
- Developed a novel algorithm to generate **Voronoi diagrams** for superquadrics and applied **Dijkstra's** algorithm on the resulting graph for efficient path planning.
- Used **image processing** to extract object **point cloud** data from an environment image, fitted superquadrics onto obstacles, and tested the algorithm on an experimental setup, using objects of various shapes as maze
- Benchmarked the Voronoi-based planner against **classical methods** like A* and RRT*, and **heuristic** planners such as genetic algorithms, and a recent state-of-the-art drone planning algorithm in 3D.
- Concluded that Voronoi planning for superquadrics offers **safer,** shorter paths with fewer nodes, **low computational space,** requiring only object **point cloud** from image, and no other parameters.

## Scholastic Achievements

2024 **Department Rank 2** out of **196** students in Mechanical Engineering, IITB

2023 Awarded the **Institute Academic Prize two times**, for exceptional academic performance in the second and third year among Mechanical Engineering students

Recipient of the **Undergraduate Research Award** for exemplary research work in the second year summer

Achieved perfect **10** Semester Performance Index in **five semesters**: Semester **3,4,5,7** and **8**

Awarded the **Academic Proficiency(AP)** grade for **exceptional academic performance** in **7 courses**

2021 Secured **All India Rank 179** in **JEE Main** among 1.2 million aspirants
Secured **All India Rank 1234** in **JEE Advanced** among 250,000 aspirants

## Projects

Feb 2026 – Present **Humanoid Motion Control for Soccer**
NomadZ – RoboCup Team, ETH Zürich | nomadz.ethz.ch

- Developing **imitation learning** based motion control for **Booster K1** humanoid robots for RoboCup soccer
- Retargeted custom-recorded human motion videos (**walking, dribbling, kicking**) onto the robot using **GVHMR** and **GMR** for human-to-robot **motion retargeting**
- Trained a **velocity-tracking locomotion policy** and a **dribbling policy** leveraging **Adversarial Motion Priors (AMP)** to encode natural style while satisfying the task objective, using **NVIDIA Isaac Lab** (MimicKit)
- Building **deployment** code for **Sim2Real** transfer and integrating the learned policies with the team's **vision** and **behaviour** pipelines

Spring 2026 **Cloth Folding with a Multi-Task Diffusion Policy**
Robot Learning Course, ETH Zürich | GitHub

- Assembled a **LeRobot SO-101** arm and collected around **200 teleoperated** demonstrations of folding a square towel in three stages: grasp a corner, first fold, and second fold
- Trained a **multi-task Diffusion Transformer (DiT)** policy (LeRobot) with **CLIP** vision and text encoders, predicting action chunks from a single wrist-camera RGB feed and language instruction
- Deployed on the **real robot** with **asynchronous inference** on a local **RTX 4060 (8GB)** – policy server at 5 FPS, robot client at 15 FPS

Dec 2025 – Jan 2026 **Monocular Visual Odometry**
Mini Project | VAMR 2025

- Built a monocular visual odometry pipeline (Python/OpenCV) using **Harris** corners + SIFT, **KLT optical flow** tracking, **PnP** + RANSAC pose estimation, triangulation to recover 6-DoF camera poses
- Added robust bootstrapping, nonlinear **pose refinement** (Levenberg–Marquardt), re-initialization, and landmark rescaling to handle drift, feature loss, and scale ambiguity
- Tuned and validated across KITTI, Malaga, Parking, and a custom ETH Polybahn ride dataset, achieving **locally consistent** trajectories. Results can be viewed here.

Aug 2024 – Apr 2025 **Koopman/DMD based Control for Non-Linear Dynamical Systems**
Bachelor Thesis Project, IIT Bombay | Guide: Prof. Abhishek Gupta

- Extensively Studied **Koopman theory**, and many data-driven control strategies, including Dynamic Mode Decomposition**(DMD** and **EDMD**), and its control variants(DMDc), and **Hankel-based MPC**
- Applied Koopman Theory on simple problems, such as spring- mass systems, vanderpol oscillators, pendulum for predicting dynamics, and applying to more complex problems such as a **two-link manipulator**
- Applied Koopman/DMD techniques on the simulation data for a planar soft robot for getting **approximate linear dynamics** in the**lifted space**

Aug 2024 – Sep 2024 **Warehouse Drone - E-Yantra, Nation-wide Robotics competition**
Competiton | eYRC portal

- Developed PID control agorithm for roll, pitch,yaw and throttle control of a Swift Pico Drone in Gazebo, with the node files and communication pipeline in ROS2 environment
- Used **A*** as the **motion planning** algorithm to **navigate** and simultaneously control the drone within a **warehouse** environment in simulation
- Team ranked **40 among 3000+ teams** across the nation participating in eYRC, and among the top 50 teams selected for the **hardware phase**, to implement our algorithms on a Quadcopter

## Positions of responsibility

May 2023 – July 2023 **Teaching Assistant | Introduction to Quantum Physics (PH107)**
Prof Alok Shukla, Department of Physics

- Conducted tutorial sessions for **40+ students** across various departments
- Covered content and clarified doubts of the students pertaining to **Quantum Physics**
- Graded the Midsemester and Endsemester answer sheets for the batch and conducted crib sessions

May 2023 – May 2024 **Student Mentor | Mechanical Department Academic Mentorship Programme**
Selected as a part of a 35 membered team out of 125+ applicants based on interviews and peer reviews

- Mentored **5 sophomores**, helping them strike a balance between academics and extracurriculars
- Acting as the first point of contact in order to actively bridge the gap between faculty and students

May 2024 – Oct 2024 **DCAMP Mentor | Mechanical Department, IIT Bombay**
An initiative by the Mechanical department to ensure proper career advice for pre-final year students

- Mentored **4 pre-final year students**, helping them get a better perspective of career opportunities
- Guided students through their internship season ensuring regular **interactions** and **advice**